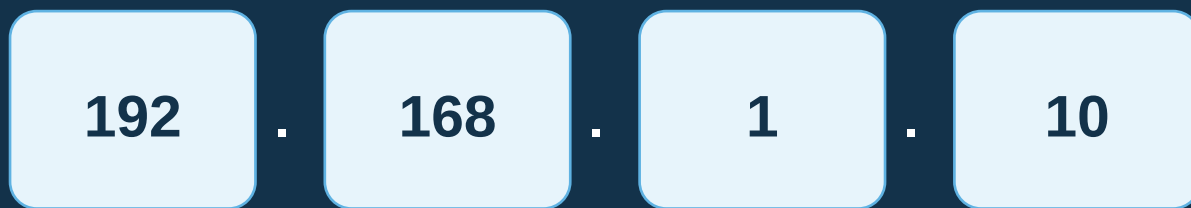


DAY 07

IPv4 Addresses

Part 1

27 original flashcards - visualized and explained



32 BITS | 4 OCTETS | CLASSFUL RANGES | PREFIXES | MASKS

What is the length of an IP address? (bits)

Four groups of 8 bits



32 bits total

ANSWER

32 bits

MEMORY CUE

IPv4 uses 32 binary bits in total.

What is the length of an IP address? (bytes)



4 bytes = 32 bits

ANSWER

4

MEMORY CUE

32 bits / 8 = 4 bytes.

What are the groups of 8 bits in an IP address called?

Octet 1

8 bits

Octet 2

8 bits

Octet 3

8 bits

Octet 4

8 bits

ANSWER

octets

MEMORY CUE

IPv4 is written as four 8-bit octets.

What is the maximum value for a binary octet? (1111 1111)

Maximum 8-bit octet



Fixed leading bits determine the historical class

11111111 = **255**

ANSWER

255

MEMORY CUE

An 8-bit unsigned value ranges from 0 to 255.

What is the numeric range of the first octet of a class A IPv4 address?

Historical classful first-octet ranges

Class A	0-127
Class B	128-191
Class C	192-223
Class D	224-239
Class E	240-255

Class A first octet = 0-127

ANSWER

0-127

MEMORY CUE

Class A first octet: 0 through 127.

What is the numeric range of the first octet of a class B IPv4 address?

Historical classful first-octet ranges

Class A	0-127
Class B	128-191
Class C	192-223
Class D	224-239
Class E	240-255

Class B first octet = 128-191

ANSWER

128-191

MEMORY CUE

Class B first octet: 128 through 191.

What is the numeric range of the first octet of a class C IPv4 address?

Historical classful first-octet ranges

Class A	0-127
Class B	128-191
Class C	192-223
Class D	224-239
Class E	240-255

Class C first octet = 192-223

ANSWER

192-223

MEMORY CUE

Class C first octet: 192 through 223.

What is the numeric range of the first octet of a class D IPv4 address?

Historical classful first-octet ranges

Class A	0-127
Class B	128-191
Class C	192-223
Class D	224-239
Class E	240-255

Class D first octet = 224-239

ANSWER

224-239

MEMORY CUE

Class D first octet: 224 through 239 (multicast).

What is the numeric range of the first octet of a class E IPv4 address?

Historical classful first-octet ranges

Class A	0-127
Class B	128-191
Class C	192-223
Class D	224-239
Class E	240-255

Class E first octet = 240-255

ANSWER

240-255

MEMORY CUE

Class E first octet: 240 through 255 (experimental/reserved).

What is the binary bit pattern of the first octet of a class A IPv4 address?

First octet bit pattern



Fixed leading bits determine the historical class

ANSWER

0xxxxxxx

MEMORY CUE

Class A begins with a leading 0 bit.

What is the binary bit pattern of the first octet of a class B IPv4 address?

First octet bit pattern



Fixed leading bits determine the historical class

ANSWER

10xxxxxx

MEMORY CUE

Class B begins with the bit prefix 10.

What is the binary bit pattern of the first octet of a class C IPv4 address?

First octet bit pattern



Fixed leading bits determine the historical class

ANSWER

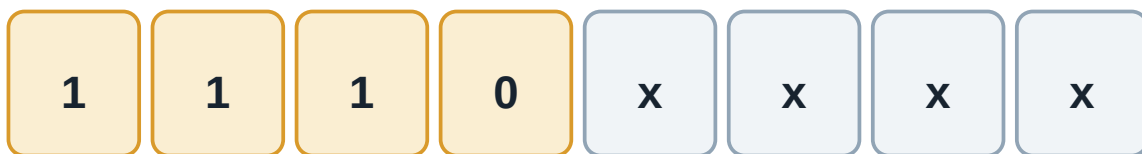
110xxxxx

MEMORY CUE

Class C begins with the bit prefix 110.

What is the binary bit pattern of the first octet of a class D IPv4 address?

First octet bit pattern



Fixed leading bits determine the historical class

ANSWER

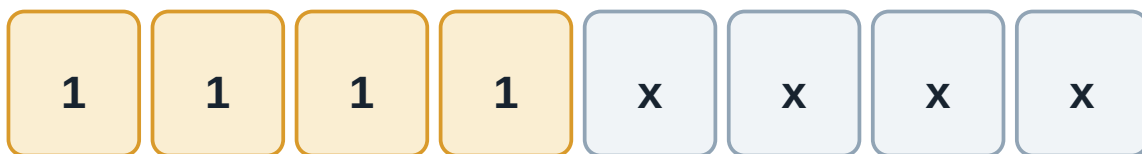
1110xxxx

MEMORY CUE

Class D begins with the bit prefix 1110.

What is the binary bit pattern of the first octet of a class E IPv4 address?

First octet bit pattern



Fixed leading bits determine the historical class

ANSWER

1111xxxx

MEMORY CUE

Class E begins with the bit prefix 1111.

What is the class D IPv4 address range reserved for?

Class D

MULTICAST

Special-use historical class range

ANSWER

Multicast addresses

MEMORY CUE

Class D addresses are used for IPv4 multicast.

What is the class E IPv4 address range reserved for?

Class E

EXPERIMENTAL

Special-use historical class range

ANSWER

Experimental purposes

MEMORY CUE

Class E is reserved for experimental purposes.

IPv4 addresses beginning with 127 are used for [...] addresses.



Traffic returns to the same host - no physical network required

ANSWER

loopback

MEMORY CUE

127.x.x.x belongs to the loopback block.

IPv4 addresses beginning with [...] are used for loopback addresses.



Traffic returns to the same host - no physical network required

ANSWER

127

MEMORY CUE

Remember loopback by the first octet: 127.

What kind of address is used to test the network software stack on the local device?



Traffic returns to the same host - no physical network required

ANSWER

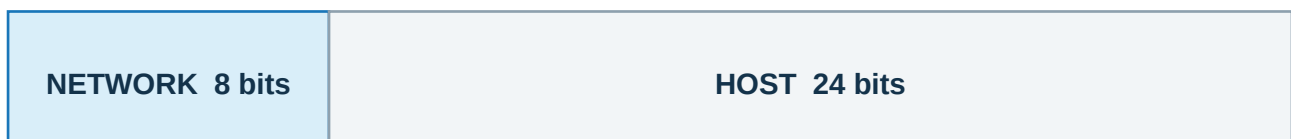
loopback address

MEMORY CUE

Loopback tests the local TCP/IP stack without leaving the device.

What is the prefix length of a class A IPv4 address?

32-bit IPv4 address



/8

ANSWER

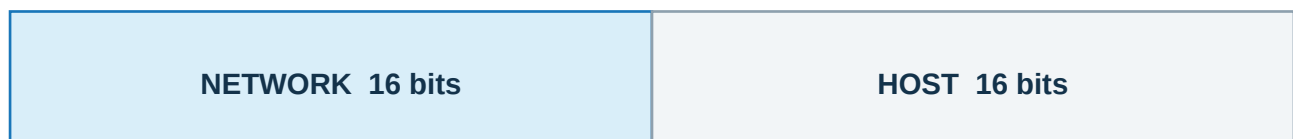
/8

MEMORY CUE

Class A uses /8 in classful addressing.

What is the prefix length of a class B IPv4 address?

32-bit IPv4 address



/16

ANSWER

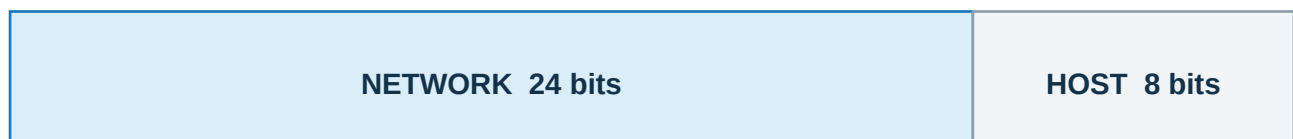
/16

MEMORY CUE

Class B uses /16 in classful addressing.

What is the prefix length of a class C IPv4 address?

32-bit IPv4 address



/24

ANSWER

/24

MEMORY CUE

Class C uses /24 in classful addressing.

What is the netmask for a /8 prefix length?



Prefix /8

ANSWER

255.0.0.0

MEMORY CUE

/8 means the first 8 mask bits are 1: 255.0.0.0.

What is the netmask for a /16 prefix length?



Prefix /16

ANSWER

255.255.0.0

MEMORY CUE

/16 means the first 16 mask bits are 1: 255.255.0.0.

What is the netmask for a /24 prefix length?



Prefix /24

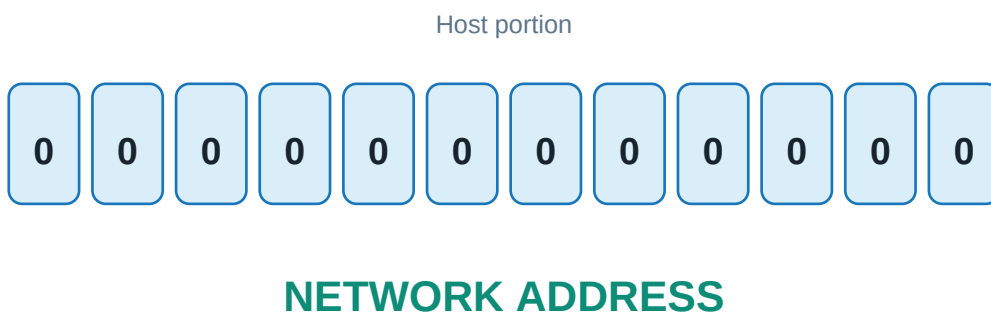
ANSWER

255.255.255.0

MEMORY CUE

/24 means the first 24 mask bits are 1: 255.255.255.0.

If the host portion of an IPv4 address is all 0s, what kind of address is it?

**ANSWER**

network address

MEMORY CUE

All host bits 0 identify the network itself.

If the host portion of an IPv4 address is all 1s, what kind of address is it?

Host portion



BROADCAST ADDRESS

ANSWER

broadcast address

MEMORY CUE

All host bits 1 identify the subnet broadcast address.

Quick Review - IPv4 Addresses (Part 1)

Class	First octet	Leading bits	Classful prefix	Purpose
A	0-127	0xxxxxxx	/8	Unicast*
B	128-191	10xxxxxx	/16	Unicast
C	192-223	110xxxxx	/24	Unicast
D	224-239	1110xxxx	-	Multicast
E	240-255	1111xxxx	-	Experimental

- IPv4 length: 32 bits = 4 bytes = 4 octets

- Loopback: 127.x.x.x; commonly 127.0.0.1

- /8 = 255.0.0.0 | /16 = 255.255.0.0 | /24 = 255.255.255.0

- Host bits all 0 -> network address

- Host bits all 1 -> broadcast address

*Classful ranges are historical addressing categories; modern networks use CIDR/prefix lengths.